

Reaching the Age of Adolescence

- **Adolescence**

- The time period when the body undergoes changes to reach reproductive maturity is known as **adolescence**.
- It begins around the age of 11 and lasts till about 18 or 19 years of age.
- Adolescence in girls can begin one or two years earlier than boys.

- **Puberty**

- The various changes that occur in the body during adolescence marks the onset of puberty.
- Puberty ends when teenagers attain sexual maturity.

- **Changes that take place during puberty**

- **Increase in height**

- It is caused by the growth in long bones of the arms and legs.
- Girls grow faster than boys initially but both reach their maximum height by the age of 18 years.

- **Change in body shape**

- Boys develop broader shoulders, wider chests, and prominent muscles.
- In girls the region below the waist becomes wider.

- **Change in the voice pattern**

- Voice box or larynx starts growing during puberty.
- It protrudes in males in the neck region and is called **Adam's apple**.
- Boys develop deep low-pitched voice.
- Girls develop high-pitched voice.

- **Change in activity of sweat and sebaceous glands**

- The activity of sweat glands increases during puberty, resulting in production of more sweat.
- The oily secretions from sebaceous glands increase. The accumulation of oil and bacterial action leads to acne problems in teenagers.

- **Changes in sex organs**

- Testes and penis develop completely in boys.
- Testes start producing sperms.
- Ovaries develop completely and start producing eggs in girls.

- **Change in intellectual level**

- The learning capacity of brain increases.
- Intellectual development takes place during adolescence.

- **Development of secondary sexual characteristics**

- Secondary sexual characteristics **in boys**

- Appearance of moustaches and beard
- Appearance of hair on chest
- Growth of hair in genital area and other parts

- Secondary sexual characteristics **in girls**

- Increase in breast size
- Growth of hair in the pubic region

- **Endocrine glands**

- Hormones are secreted by endocrine glands such as the pituitary gland, thyroid gland, adrenal gland, pancreas etc.

- Major endocrine glands in humans are

- Pituitary
- Hypothalamus
- Pineal
- Thyroid
- Parathyroid
- Thymus
- Pancreas
- Adrenal
- Testis in men /ovary in women

- A feedback mechanism (positive and negative) regulates the action of the hormones

- **Hormones**

- Hormones are chemical secretions that bring about various changes in the body
- They are produced by **endocrine glands**.
- These glands release hormones into blood to reach specific **target site**.
- Production of hormones is under the control of hormones produced from **pituitary gland**.

- **Characteristics of hormones:**

- Hormones act as chemical messengers.

- They are secreted by living cells/tissues or organs called glands.
- They are secreted in very small quantities by glands.
- They act upon specific cells, tissues, or organs called the target sites.
- They are generally slow in action, but have long lasting effects.
- They either accelerate or inhibit a reaction.

- **Various Glands**

- **Pituitary gland**

- It is a pea sized gland situated at the base of the brain. It secretes a **growth hormone (GH)**.
- It is required for proper body growth.
- The hyposecretion of growth hormone causes a condition called **dwarfism**.
- The hypersecretion causes **gigantism** in children and **acromegaly** in adults.

- **Thyroid gland**

- It is located close to trachea in the neck. It produces a hormone called **thyroxine**.
- It is required for regulating metabolism in the body.
- The hyposecretion of thyroxine causes **hypothyroidism**. This condition causes abnormalities like **simple goitre**, **myxoedema** and **cretinism**. Lack of iodine leads to deficiency of thyroxine, which results in a disease called goitre.
- The excess secretion of thyroxine causes **hyperthyroidism**. It results in high metabolism, protrusion of the eye balls, high BP, nervous tension, etc.

- **Parathyroid Gland**

- There are four parathyroid glands present on back side of thyroid glands that secrete parathyroid hormone or parathormone (PTH).
- This hormone regulates the level of calcium ions in the bloodstream.
- Excess of parathyroid hormone removes calcium from bones and makes them soft.

- **Pancreas**

- It produces two hormones- Insulin and Glucagon.
- These hormones maintain blood sugar level.
- Deficiency of insulin results in diabetes.

- **Adrenal Gland**

- There are two adrenal glands located one on upper part of each kidney.

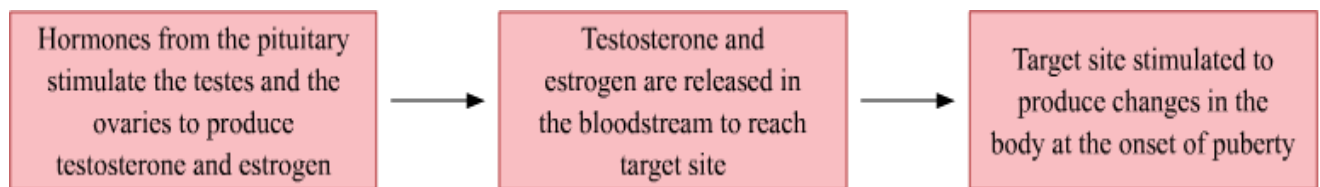
- It has two parts- **cortex** and **medulla**.
- Cortex secretes the hormones like **cortisol** that regulates the rate of metabolism.
- The medulla secretes a hormone like **adrenaline** that prepares the body to face various stressful situations.

- **Gonads**

- It includes testes in males and ovaries in females.
- Male sex hormone is **testosterone**. It is produced by the testes on the onset of puberty.
- Female sex hormones produced by ovaries are **estrogen** and **progesterone**.
- Deficiency of estrogen causes **infertility**.

Process of Hormonal Action

- Endocrine glands release their secretions (**hormones**) into the bloodstream.
- Hormones, on reaching their **target site**, bring about necessary changes to maintain proper functioning of the body.
- Let us take examples of hormones, **testosterone** and **estrogen**, to understand the process of hormonal action.



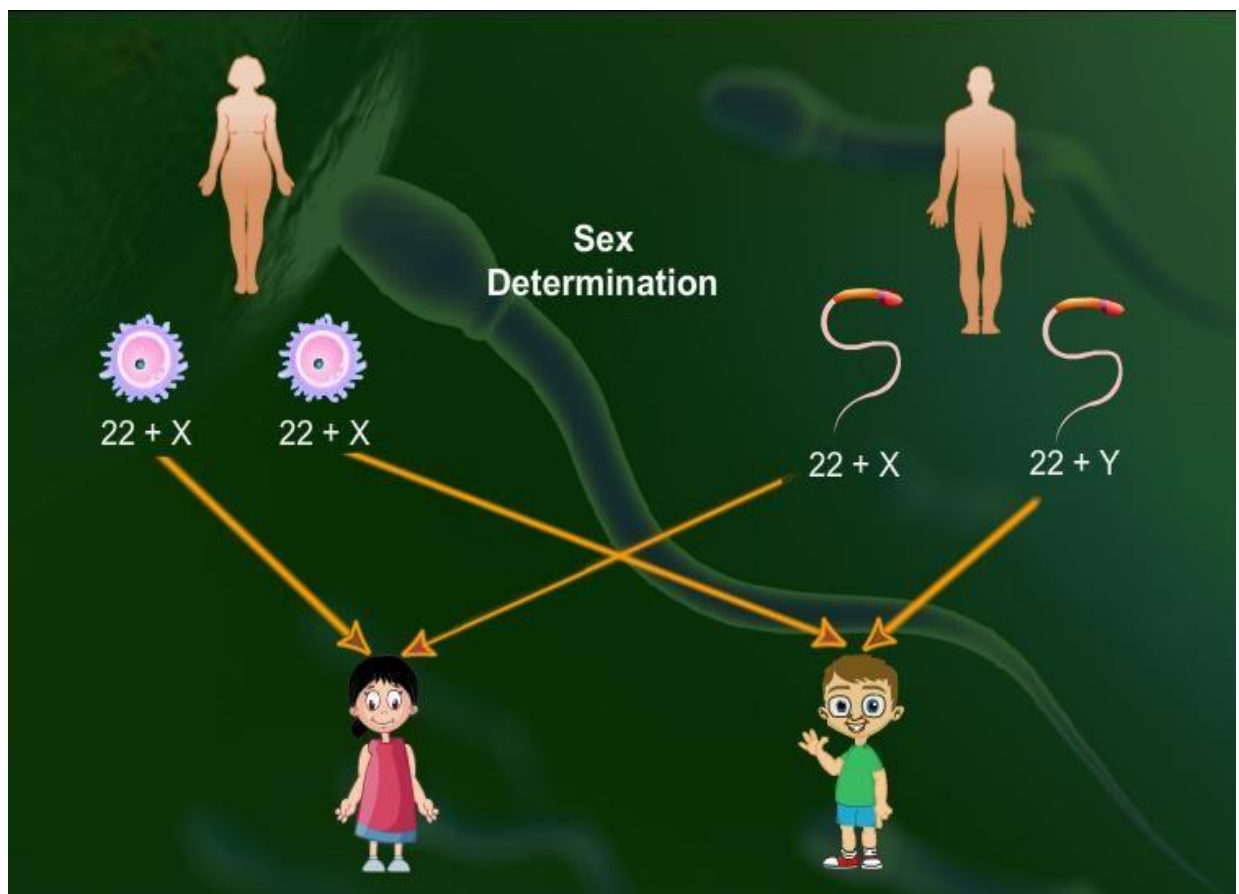
- **Reproductive phase in humans**

- In males, the production of sperm continues throughout the life.
- In females, the reproductive phase starts from 10-12 years and continues till 45-50 years.
- The female reproductive tract undergoes series of cyclic changes, called menstrual cycle, which is of 28 to 30 days.
- One ovum is produced during one cycle.
- The wall of uterus becomes thick to receive fertilized egg.
- If pregnancy does not occur, the unfertilized egg and uterus lining are shed off, which results in bleeding, called **menstruation**.

- First menstrual flow in a female is called **menarche**.
- Stoppage of menstruation in females is called **menopause**.

Sex determination in humans

- Autosomes- first 22 pairs of chromosomes that do not determine the sex of an individual.
- Sex chromosomes- last pair of chromosomes, represented as X and Y.
- Females have two X chromosomes, so can be represented as 44+XX.
- Males have one X and one Y chromosome, so can be represented as 44+XY.



- Each gamete receives half of the chromosomes i.e. 22+X or 22+Y.
- Male gametes have 22 autosomes and either X or Y sex chromosome.
- Male gametes can be of two types, 22+X or 22+Y.
- Female gametes can be of only one type, 22+X.

- Sex of a baby is determined by the type of the male gamete (X or Y) that fuses with the female gamete.
- **Personal health and hygiene in adolescents**
 - Adolescents should have a balanced diet with right proportions of various nutrients.
 - Adolescents should maintain cleanliness to prevent bacterial infections.
 - They should indulge in some physical exercises to keep their bodies fit.
 - They should avoid the consumption of drugs and alcohol.
- **Historical background and Discovery of HIV**
 - The first cases of AIDS were recognized in U.S.A in the year **1981**.
 - The AIDS virus was first discovered by the team of French scientists lead by **Luc Montagnier in 1983**.
 - In **1984**, the American virologist named **Robert Charles Gallo** gave the first report on the virus causing AIDS.
 - The name HIV was suggested by the **International committee on the nomenclature of viruses**.
 - In **India** the firsts AIDS patient was identified in **Chennai** in the year **1987**.
- **AIDS or Acquired Immune deficiency syndrome** is a viral disease, caused by the deadly virus (**HIV**).
- **Structure of HIV**
 - Human Immunodeficiency virus (HIV) is spherical in shape and contains RNA as its genetic material.
 - Externally, the virus is covered by the double layered membrane made up of fatty substances.
 - Inside the fatty membrane a core of proteins is found that surrounds the viral RNA along with the enzyme reverse transcriptase.
- **Transmission:**
 - Sharing of syringes during drug abuse.
 - Unsafe sexual contact.

- Transfusion of infected blood
- From infected mother to her infant through milk.

- **Prevention**

- Avoid sexual contact with infected persons
- Ensure use of disposable syringes
- Screening blood from blood banks

- **Tests for detection**

- **PCR** (Polymerase Chain Reaction)
- **ELISA** (Enzyme linked Immuno Sorbent Assay)
- **Western Blot**